

# Report

|                             | flye               | hifiasm            | lja                |
|-----------------------------|--------------------|--------------------|--------------------|
| # contigs (>= 0 bp)         | 62                 | 514                | 332                |
| # contigs (>= 1000 bp)      | 58                 | 514                | 332                |
| # contigs (>= 5000 bp)      | 57                 | 514                | 237                |
| # contigs (>= 10000 bp)     | 55                 | 514                | 158                |
| # contigs (>= 25000 bp)     | 47                 | 494                | 34                 |
| # contigs (>= 50000 bp)     | 46                 | 106                | 21                 |
| Total length (>= 0 bp)      | 141728081          | 161727857          | 145958069          |
| Total length (>= 1000 bp)   | 141725691          | 161727857          | 145958069          |
| Total length (>= 5000 bp)   | 141721502          | 161727857          | 145701658          |
| Total length (>= 10000 bp)  | 141703407          | 161727857          | 145081403          |
| Total length (>= 25000 bp)  | 141573733          | 161288482          | 143162763          |
| Total length (>= 50000 bp)  | 141548557          | 146846450          | 142754430          |
| # contigs                   | 62                 | 514                | 332                |
| Largest contig              | 13978706           | 33625263           | 33622357           |
| Total length                | 141728081          | 161727857          | 145958069          |
| Reference length            | 119667750          | 119667750          | 119667750          |
| GC (%)                      | 36.41              | 36.58              | 36.71              |
| Reference GC (%)            | 36.06              | 36.06              | 36.06              |
| N50                         | 5635704            | 9991083            | 11384185           |
| NG50                        | 6618624            | 11385964           | 13493880           |
| N90                         | 1773478            | 54286              | 4035433            |
| NG90                        | 3091494            | 6618975            | 8610758            |
| auN                         | 6412526.4          | 13982571.9         | 16002232.4         |
| auNG                        | 7594653.2          | 18897082.8         | 19517831.2         |
| L50                         | 9                  | 5                  | 4                  |
| LG50                        | 7                  | 3                  | 3                  |
| L90                         | 25                 | 82                 | 12                 |
| LG90                        | 17                 | 8                  | 7                  |
| # misassemblies             | 4295               | 5388               | 5319               |
| # misassembled contigs      | 40                 | 166                | 112                |
| Misassembled contigs length | 119504816          | 135212554          | 131403311          |
| # local misassemblies       | 2410               | 2911               | 3031               |
| # scaffold gap ext. mis.    | 0                  | 0                  | 0                  |
| # scaffold gap loc. mis.    | 0                  | 0                  | 0                  |
| # unaligned mis. contigs    | 15                 | 2                  | 2                  |
| # unaligned contigs         | 0 + 47 part        | 0 + 22 part        | 0 + 23 part        |
| Unaligned length            | 27218165           | 26054362           | 26169130           |
| Genome fraction (%)         | 90.483             | 88.903             | 90.605             |
| Duplication ratio           | 1.057              | 1.271              | 1.101              |
| # N's per 100 kbp           | 0.00               | 0.00               | 0.00               |
| # mismatches per 100 kbp    | 756.02             | 678.03             | 777.65             |
| # indels per 100 kbp        | 152.88             | 141.26             | 150.68             |
| # genomic features          | 767988 + 7086 part | 752115 + 6837 part | 768600 + 7067 part |
| Largest alignment           | 754365             | 754365             | 754374             |
| Total aligned length        | 114265701          | 135040360          | 119219896          |
| NA50                        | 62113              | 48271              | 58734              |
| NGA50                       | 87257              | 85900              | 87280              |
| NA90                        | -                  | -                  | -                  |
| NGA90                       | 5550               | 26221              | 7805               |
| auNA                        | 108855.6           | 99883.1            | 106587.4           |
| auNGA                       | 128922.8           | 134989.4           | 130004.0           |
| LA50                        | 496                | 681                | 529                |
| LGA50                       | 347                | 349                | 346                |
| LA90                        | -                  | -                  | -                  |
| LGA90                       | 2274               | 1420               | 2111               |

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

## Misassemblies report

|                             | flye      | hifiasm   | lja       |
|-----------------------------|-----------|-----------|-----------|
| # misassemblies             | 4295      | 5388      | 5319      |
| # contig misassemblies      | 4295      | 5388      | 5319      |
| # c. relocations            | 2690      | 3625      | 3514      |
| # c. translocations         | 1580      | 1738      | 1781      |
| # c. inversions             | 25        | 25        | 24        |
| # scaffold misassemblies    | 0         | 0         | 0         |
| # s. relocations            | 0         | 0         | 0         |
| # s. translocations         | 0         | 0         | 0         |
| # s. inversions             | 0         | 0         | 0         |
| # misassembled contigs      | 40        | 166       | 112       |
| Misassembled contigs length | 119504816 | 135212554 | 131403311 |
| # local misassemblies       | 2410      | 2911      | 3031      |
| # scaffold gap ext. mis.    | 0         | 0         | 0         |
| # scaffold gap loc. mis.    | 0         | 0         | 0         |
| # unaligned mis. contigs    | 15        | 2         | 2         |
| # mismatches                | 863875    | 915615    | 927109    |
| # indels                    | 174690    | 190759    | 179644    |
| # indels (<= 5 bp)          | 142086    | 156169    | 145806    |
| # indels (> 5 bp)           | 32604     | 34590     | 33838     |
| Indels length               | 977232    | 1053034   | 1069345   |

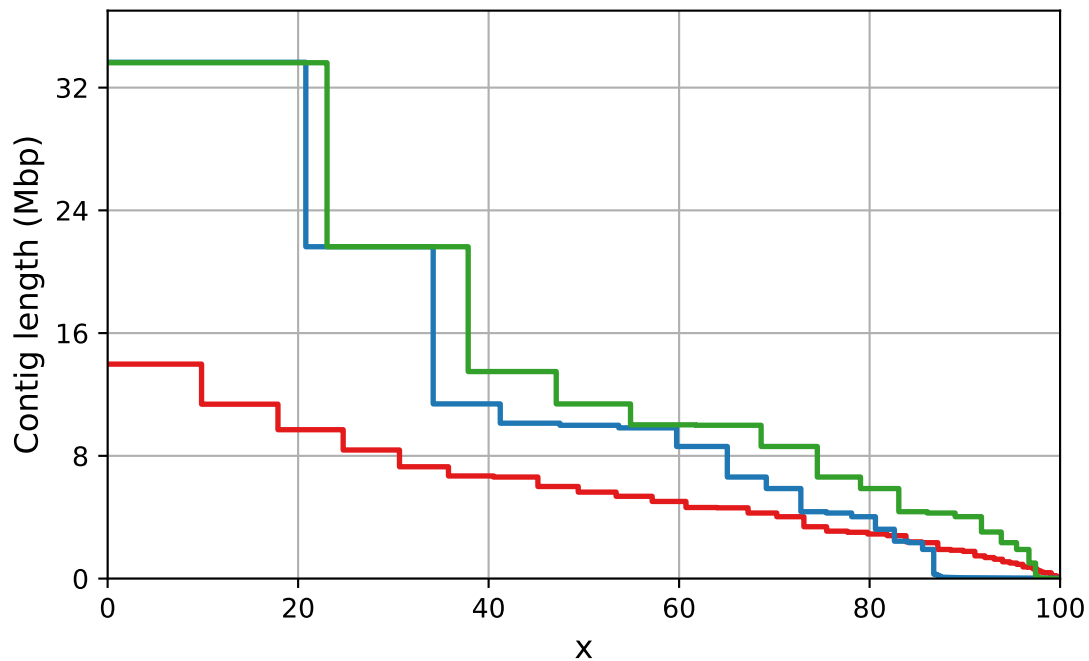
All statistics are based on contigs of size  $\geq 500$  bp, unless otherwise noted (e.g., "# contigs ( $\geq 0$  bp)" and "Total length ( $\geq 0$  bp)" include all contigs).

## Unaligned report

|                               | flye     | hifiasm  | lja      |
|-------------------------------|----------|----------|----------|
| # fully unaligned contigs     | 0        | 0        | 0        |
| Fully unaligned length        | 0        | 0        | 0        |
| # partially unaligned contigs | 47       | 22       | 23       |
| Partially unaligned length    | 27218165 | 26054362 | 26169130 |
| # N's                         | 0        | 0        | 0        |

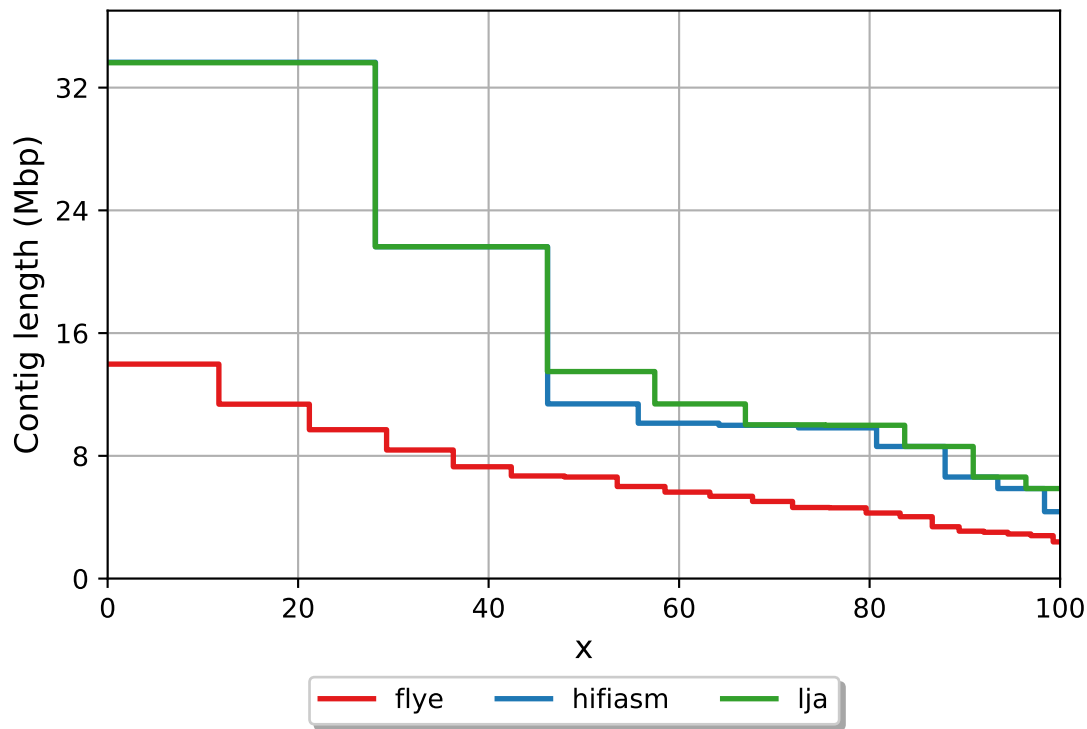
All statistics are based on contigs of size  $\geq 500$  bp, unless otherwise noted (e.g., "# contigs ( $\geq 0$  bp)" and "Total length ( $\geq 0$  bp)" include all contigs).

Nx

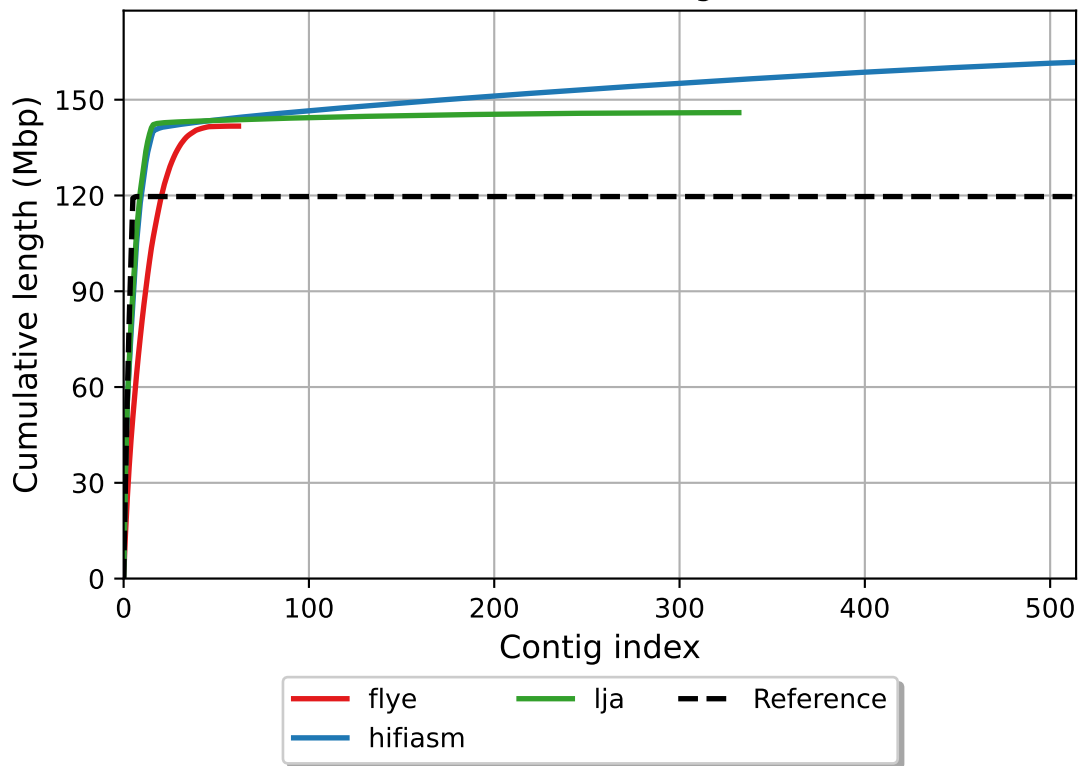


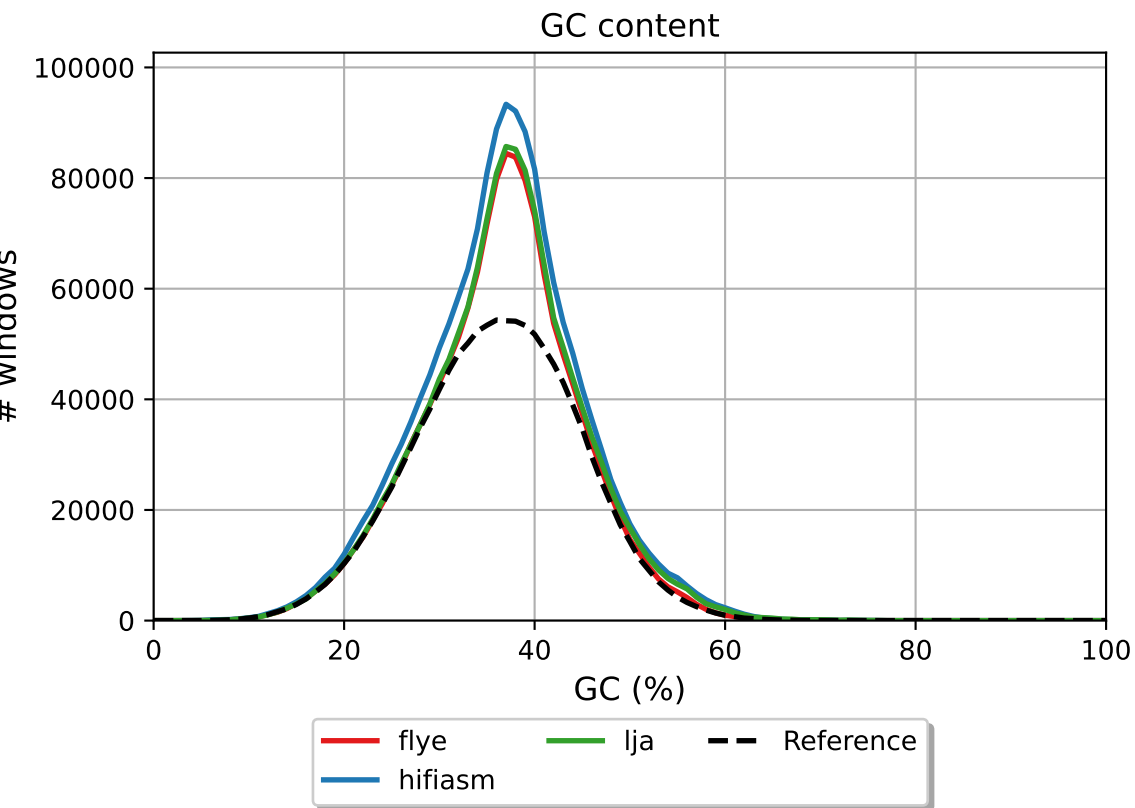
flye hifiasm lja

# NGx

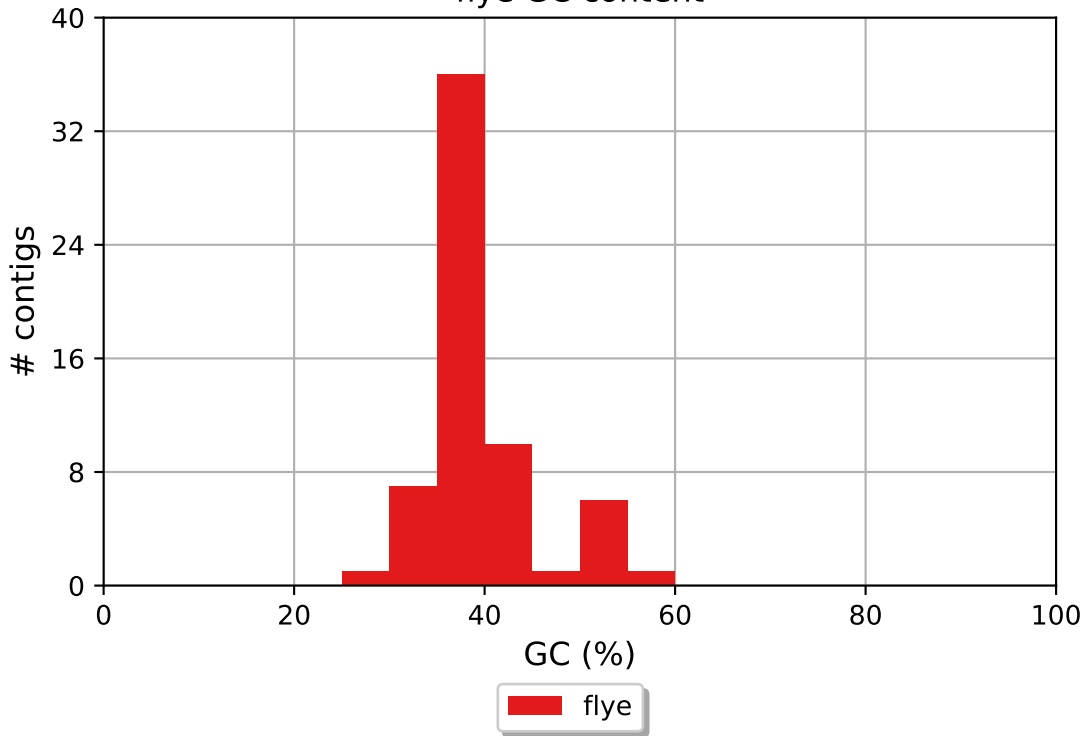


Cumulative length

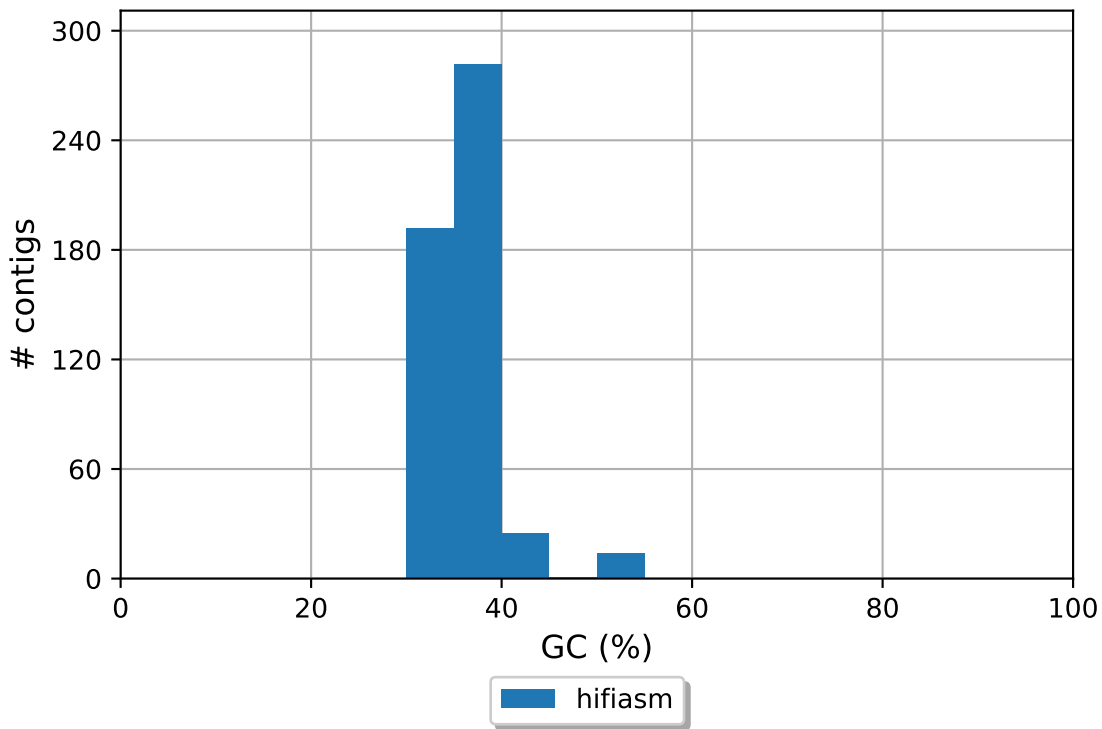




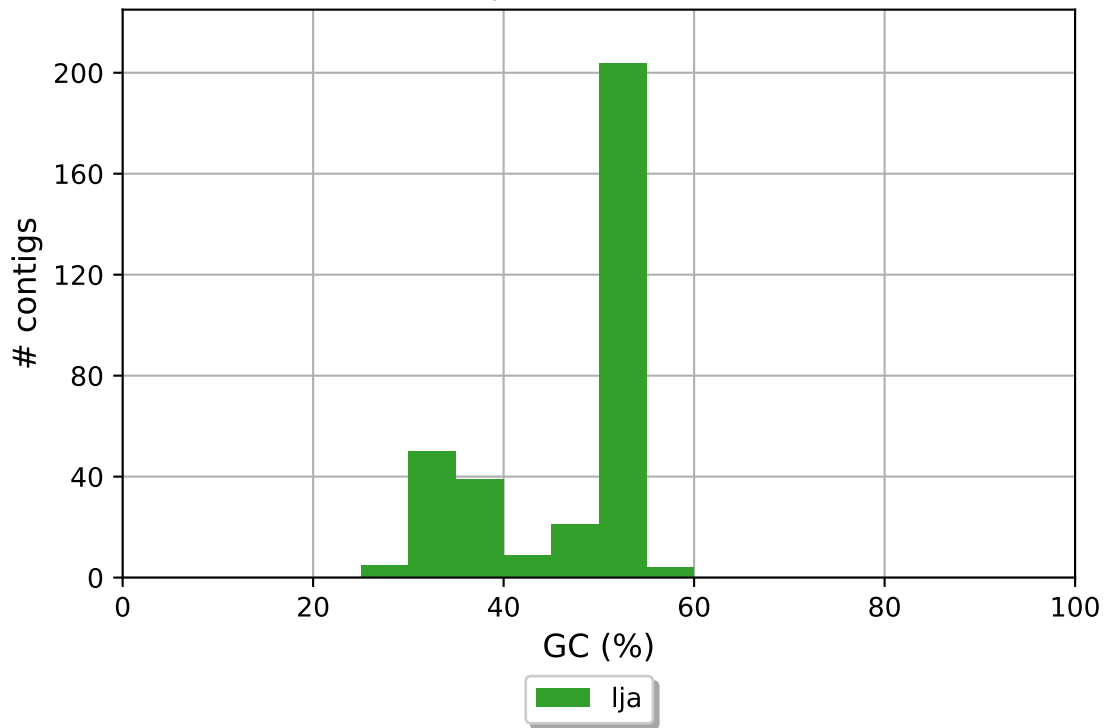
flye GC content



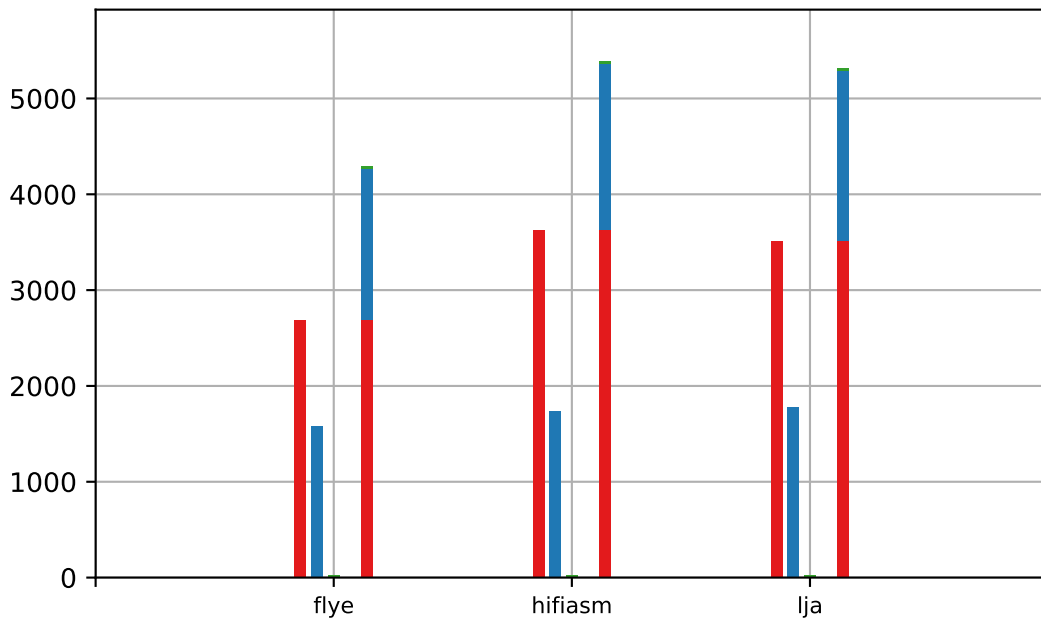
hifiasm GC content



lja GC content

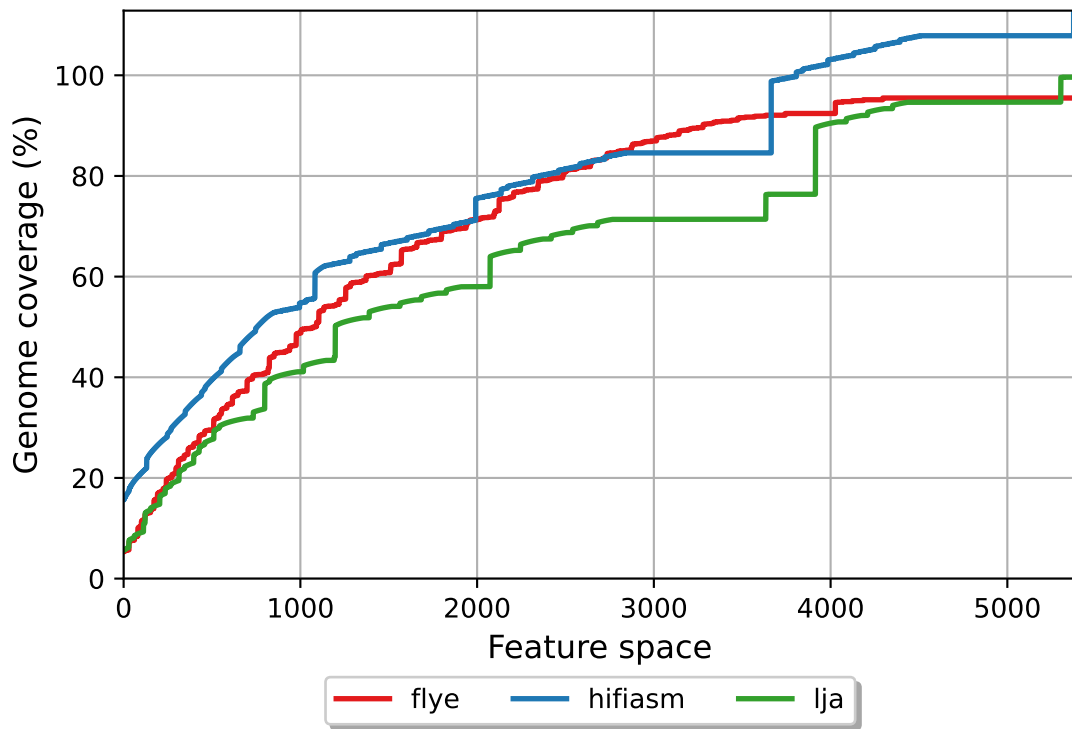


## Misassemblies

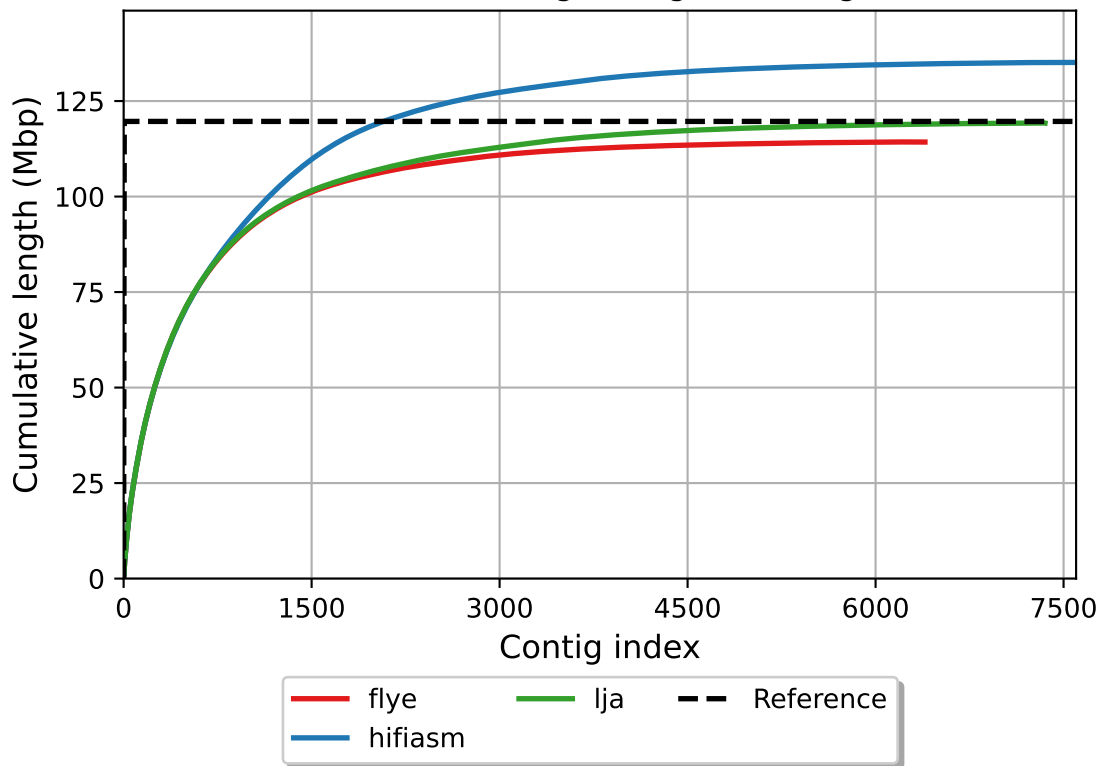


■ # relocations    ■ # translocations    ■ # inversions

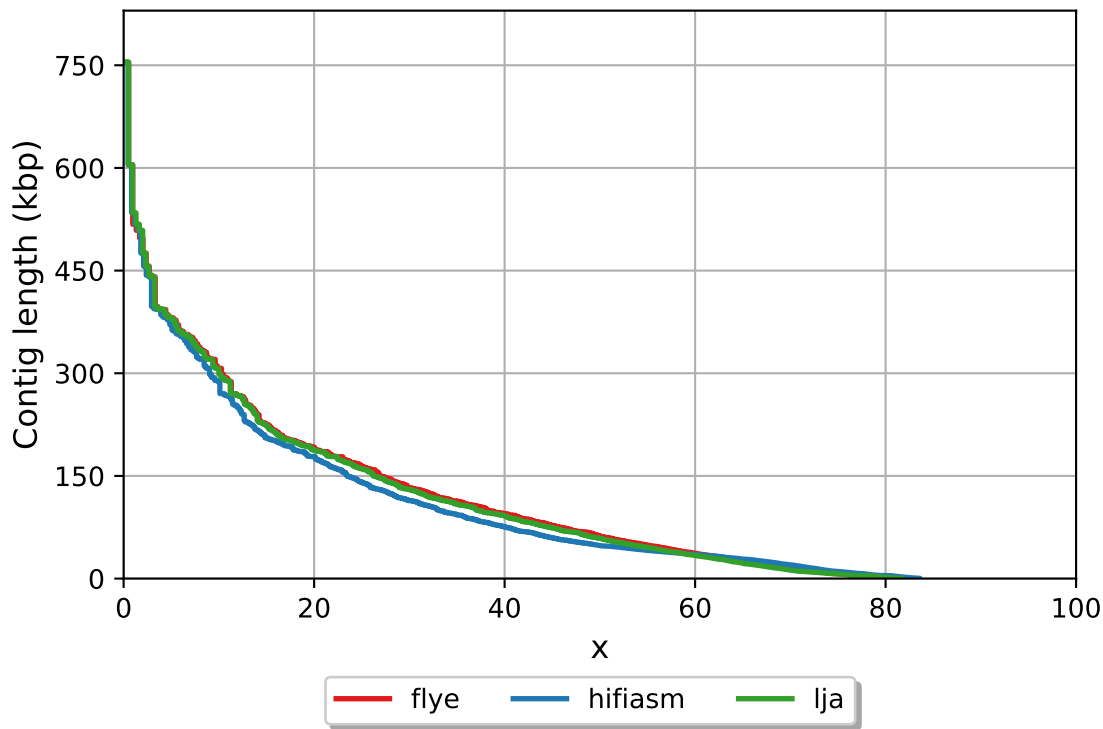
FRCurve (misassemblies)



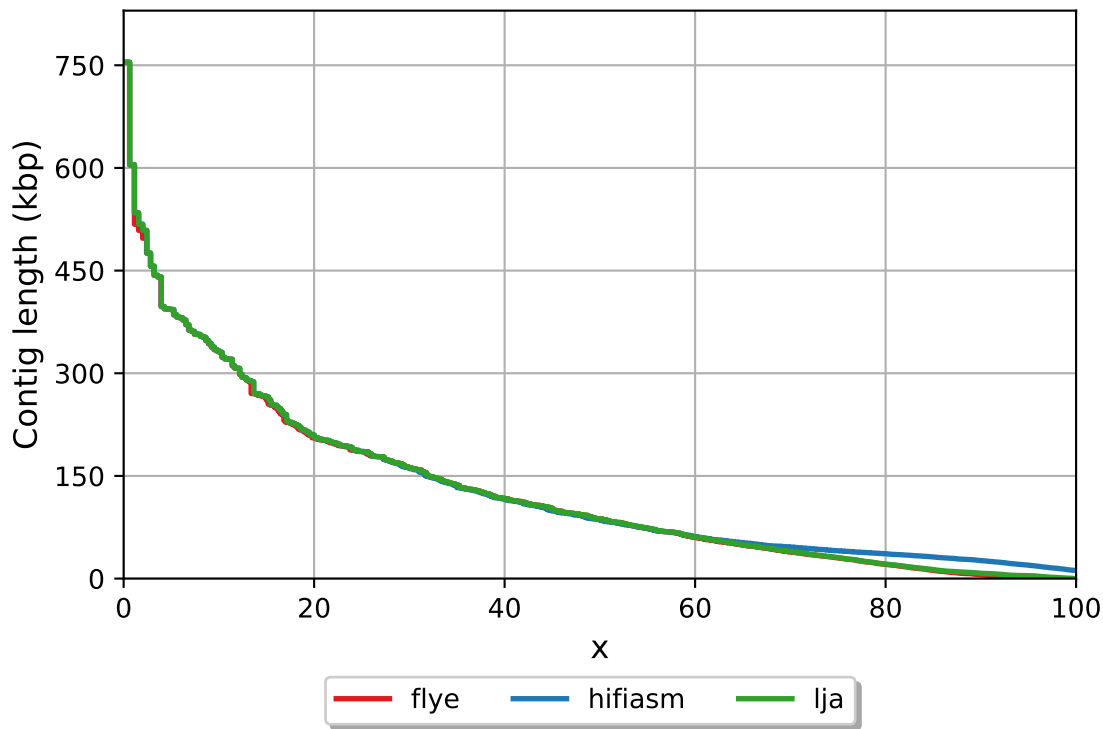
Cumulative length (aligned contigs)



# NAx

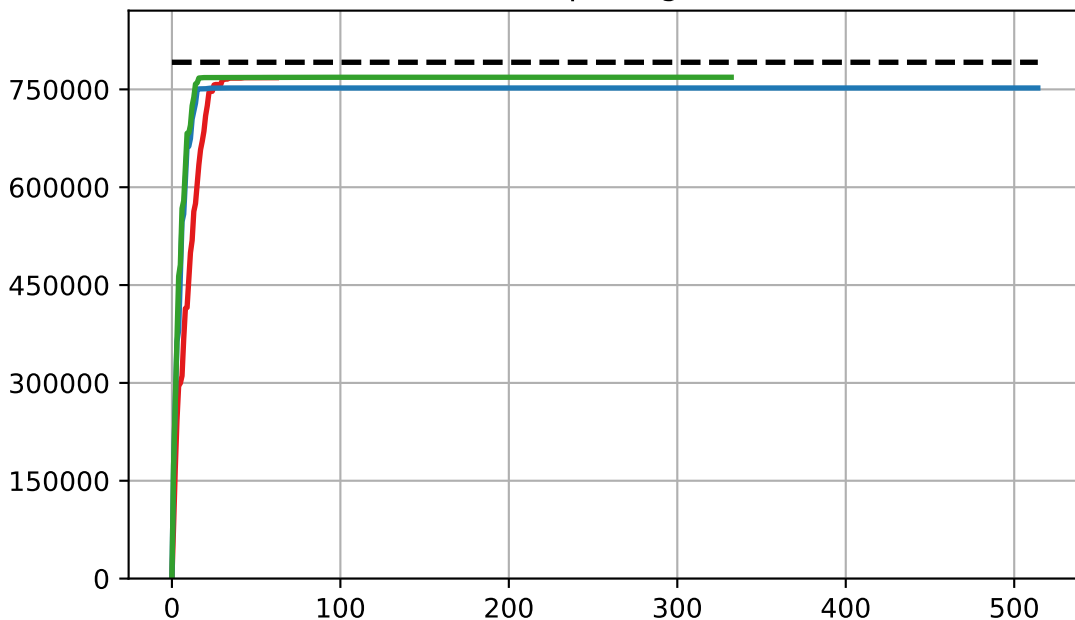


# NGAx



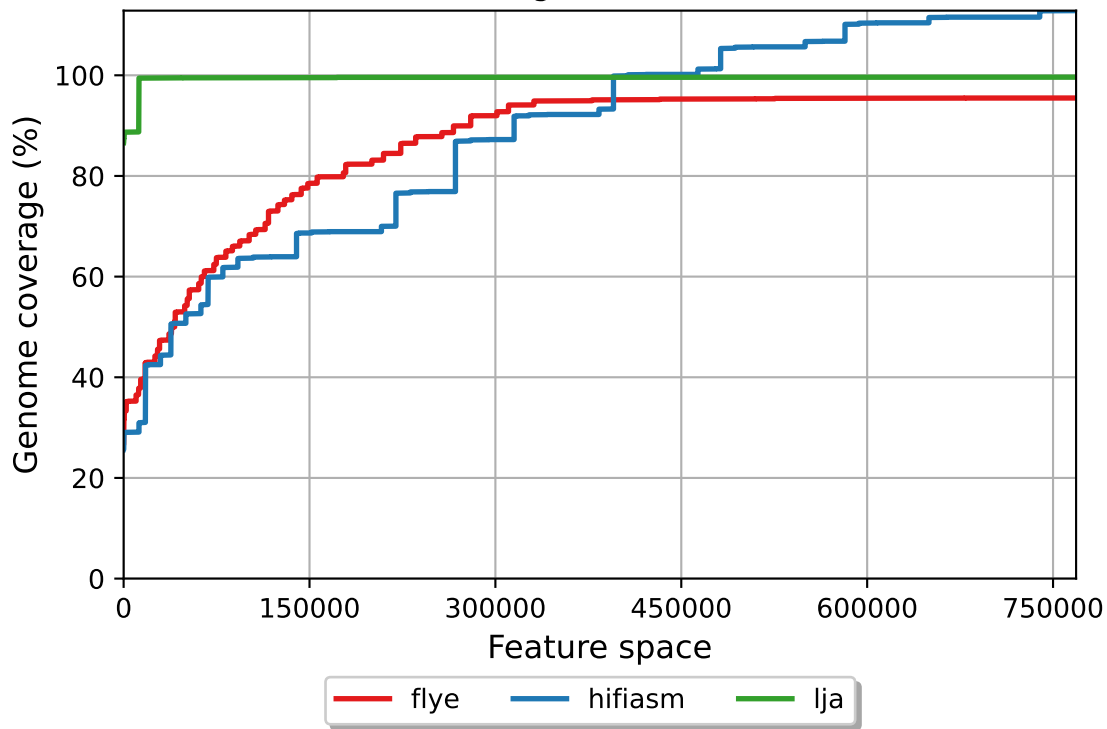
Cumulative # complete genomic features

Cumulative # complete genomic features

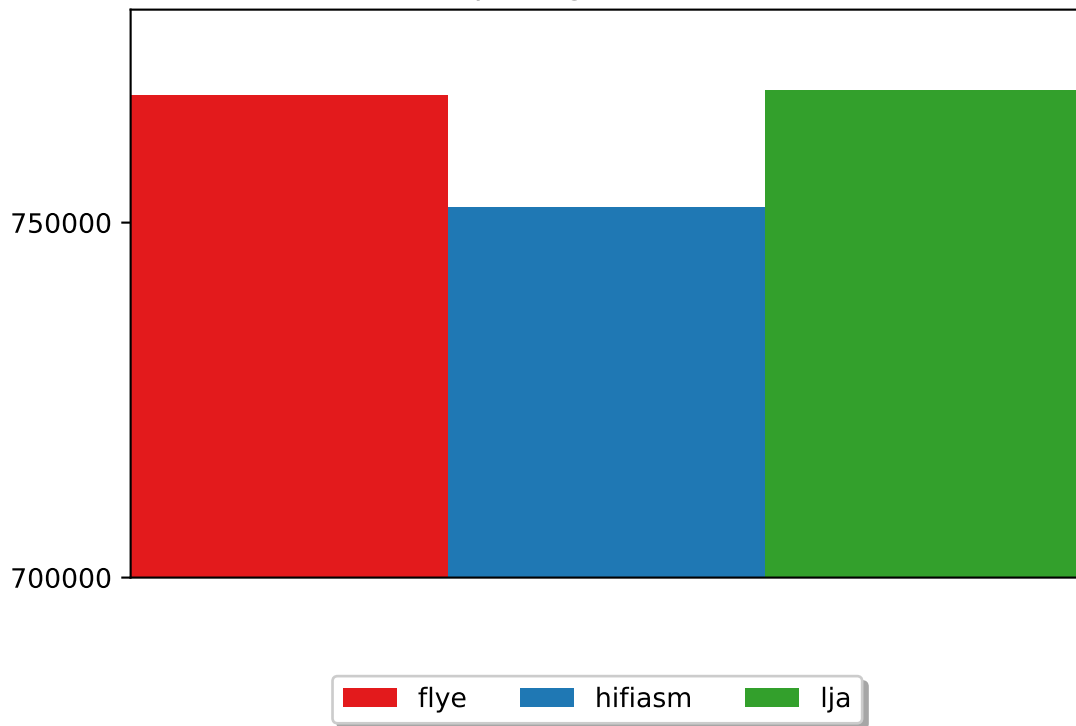


flye lja Reference  
hifiasm

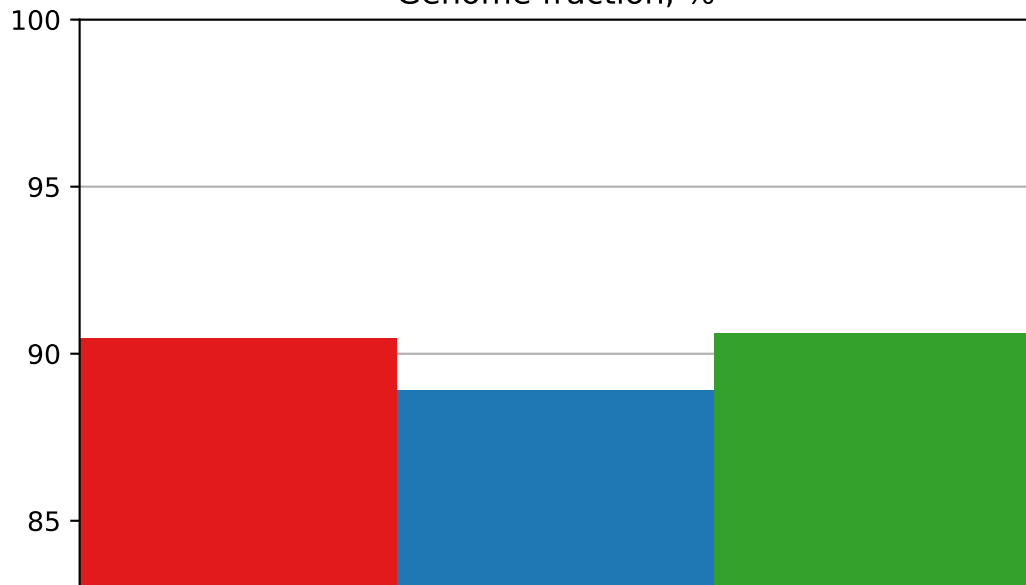
FRCurve (genomic features)



# complete genomic features



Genome fraction, %



flye hifiasm lja